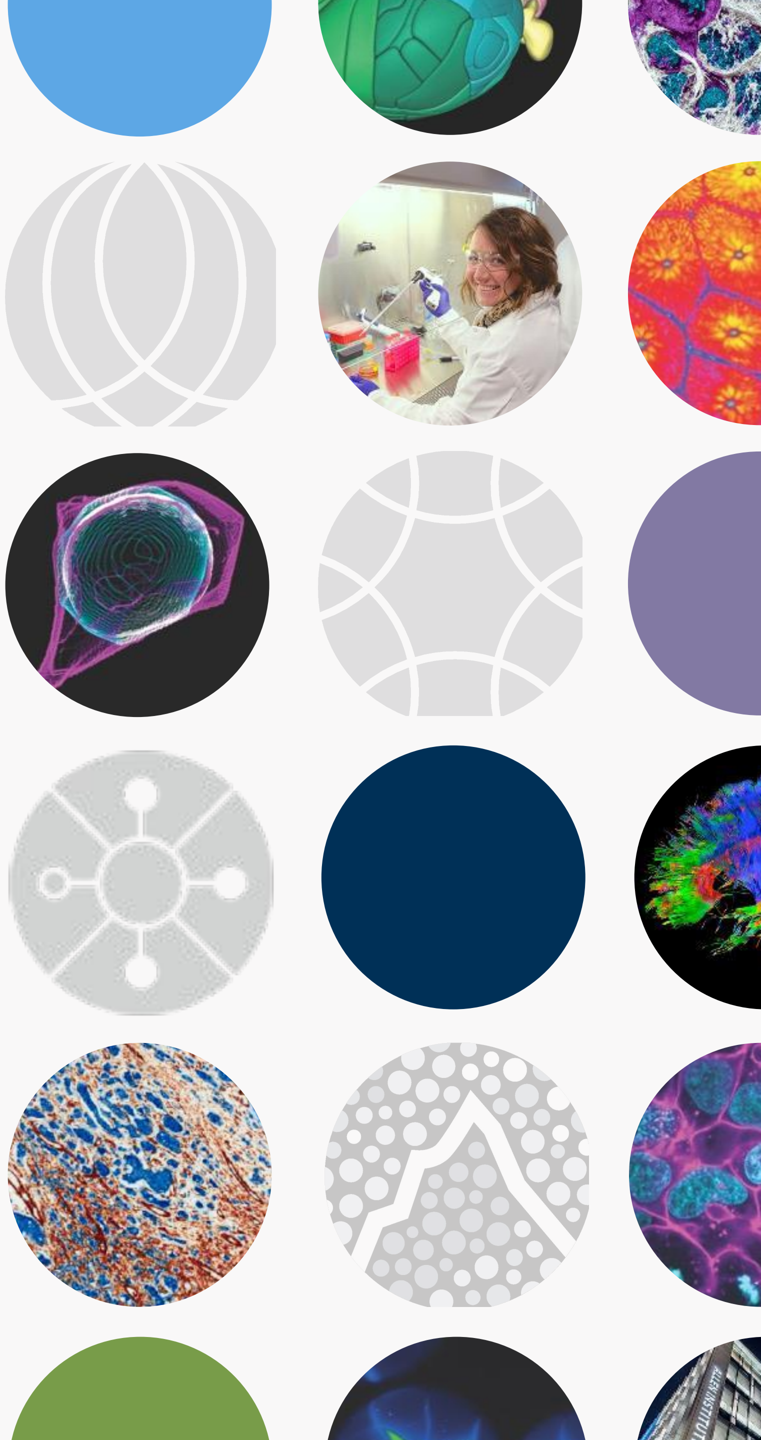
Making this your own



Learning objectives for this morning

- Address your outstanding questions
- Introduce various platforms for implementing Python coding & additional resources to learn more
- Identify when you need to involve campus IT or not
- Compare & contract choices in creating and sharing your programming lesson plans
- Introduce JupyterBook as a way to stitch together separate notebooks
- Generate your plan!

Your requests!

Available datasets

Ashley's semi-curated list: http://bit.ly/openneurodatasets

There is also a spreadsheet of Allen datasets in the workshop folder 😎
https://docs.google.com/spreadsheets/d/1xN7y1cRM4C64V7Ygos8DDBSI1Omo2LKd32z9kaHjMoQ/edit?usp=sharing

- You saw how to pull from `allensdk` with Saskia's lesson
- We'll see one more example in Cell Types in a moment

Online resources: Finding datasets

Tons of large, open access datasets, with some examples*:

- https://openneuro.org/
- https://crcns.org/data-sets
- https://www.humanconnectome.org/
- https://www.nitrc.org/
- https://osf.io/
- https://openlists.github.io/
- https://www.nwb.org/example-datasets/

* Not an exhaustive list

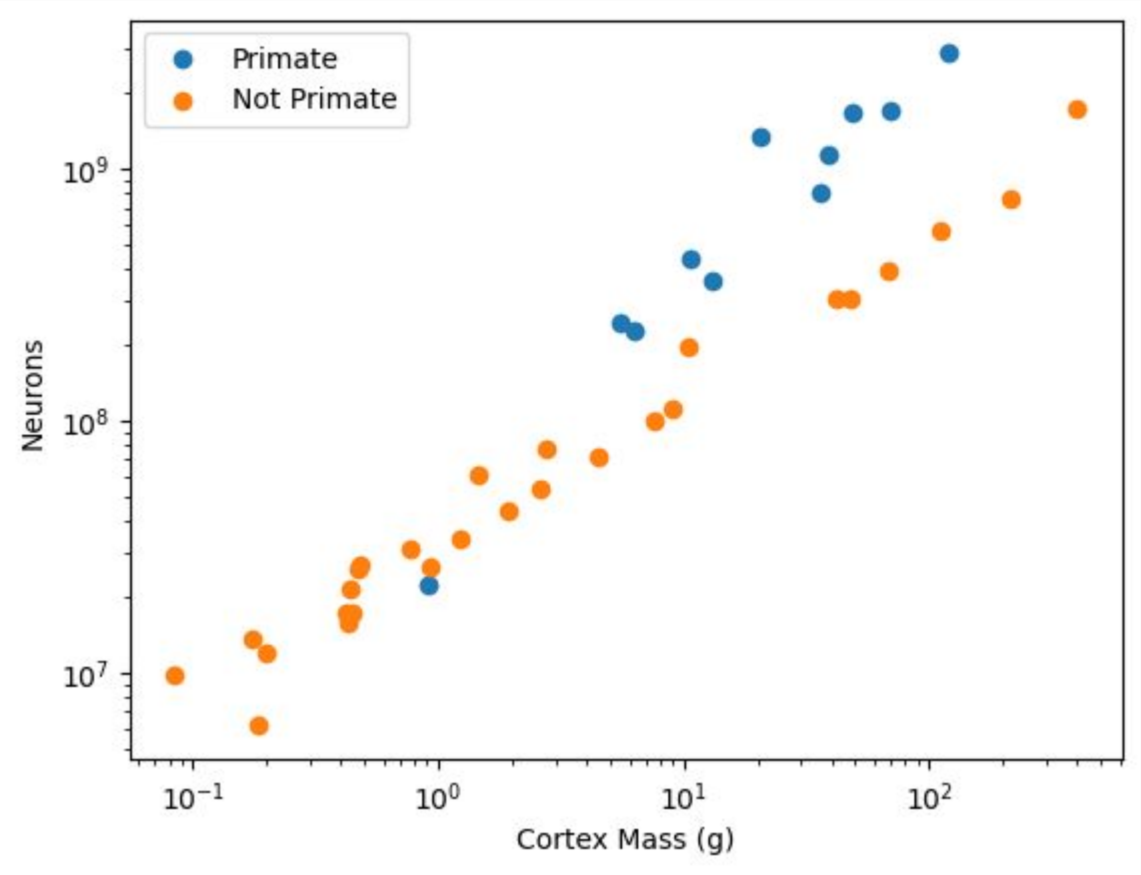
Slide: Theresa McKim

Datasets that are simple enough for teaching

Species vs. Brainmass

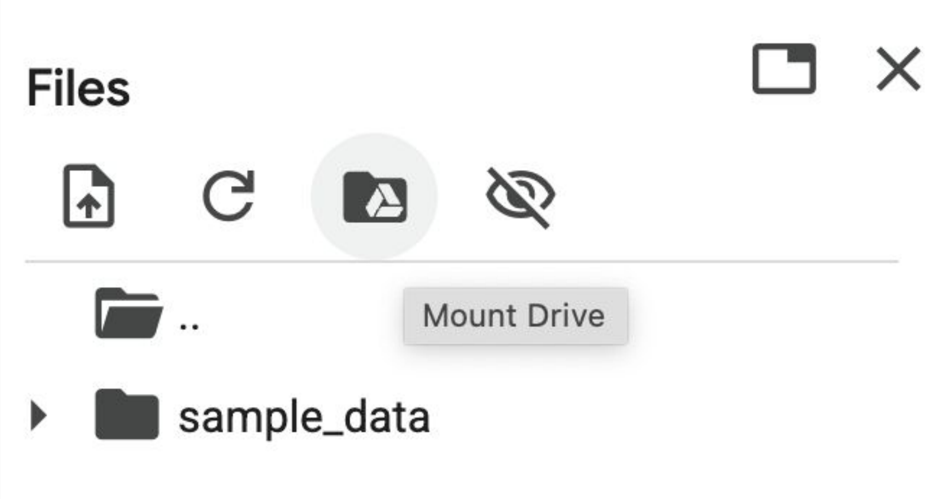
[Notebook][Data]

Allen Institute Education Materials Library

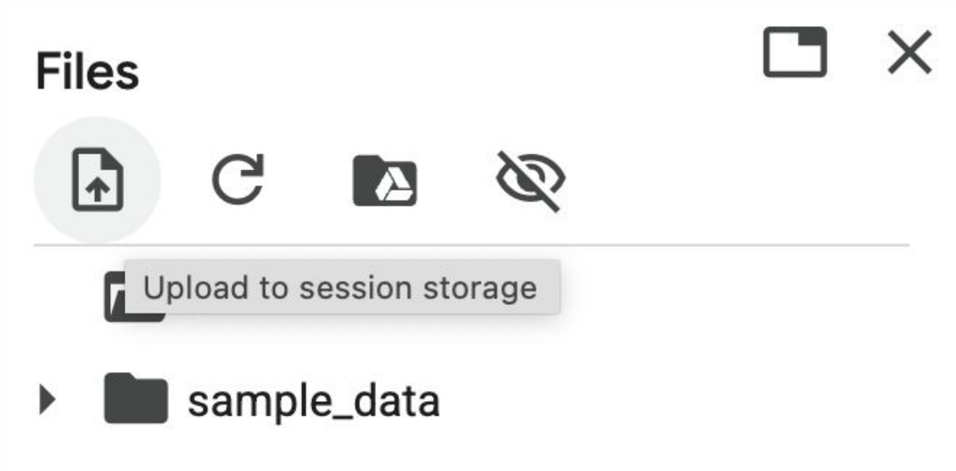


Uploading your own data to Colab

Put files in Google and "mount" Google Drive



Students upload files to Colab:



[Video Explainer]

Uploading your own data to Colab

Kinds of data you might want to import:

- Excel spreadsheet → comma-separated (`.csv`) or tab-separated (`.tsv`)
- Text files (`.txt`)
- MATLAB files (`.mat`)
- Images?

Struggles and rewards

What do students struggle with the most?

- Getting started, early frustrations
- Arrays
- Troubleshooting errors

… a lot of the same things you did!

What prompts/exercises are most engaging?

- Visualizing data
- Student decision-making & agency

… a lot of the same things you did!

How to start your own notebook

and vibe code.

Integrated Development Environments (IDEs)

- Help you write, debug, and compile code
  - **Compiling** is the process of translating your **source code** into **machine code**
- Useful because they have features like **line numbers** and **syntax highlighting**, which colors your code based on the syntax.
- Often have auto-completion, memory for commands, and provide information about functions

Visual Studio (VS Code)

- Supports many different languages, highly customizable
- Integrates with GitHub Copilot

Link:

https://code.visualstudio.com/download

Anaconda is an open-source distribution of Python, focused on scientific computing in Python.

Includes:

- “Conda,” a package management tool
- Useful code packages
- A couple applications for editing & running code:
    - Spyder (Python IDE)
    - Jupyter Notebooks

ANACONDA®

Talking to IT

- If Google Colab won't work for your purposes, you may want to set up / use a JupyterHub (e.g. http://datahub.ucsd.edu/)
- You can consider using Docker "containers" to make sure you have a reproducible computing environment (otherwise you're subject to the whims of Colab updates!)
  - Likely not necessary for you, but good to know about!
  - Local runtimes - Google Colab

Integrating what we've learned into your teaching

	Exposure to coding in coursework	Integration of coding in coursework	Discipline-based coding classes
Learning Outcomes *Students will be able to:*	• Recognize use cases for programming in neuroscience • Edit provided code examples, often to generate figures or analyze data	*Previous column, plus:* • Write additional code based on provided examples • Edit code to address a research question	*Previous column, plus:* • Design coding workflows to address a data analysis challenge *and/or* • Write code to illustrate a computational model
Time required	< 1 hour	> 1 hour	One semester/ quarter
Instructor comfort with coding	Novice: Instructors can troubleshoot basics of coding notebooks and simple examples	Novice: Instructors can troubleshoot more complex code examples	Expert: Instructors can explain mechanics of code, troubleshoot complex code, write additional examples and problem sets for students, and write code to analyze neuroscience data
Examples	• Implementation in a laboratory course: https://bipn145.github.io/	• Analysis of Allen Institute cell types data: Juavinett, 2020; http://github.com/ajuavinett/CellTypesLesson/ and Ho et al., 2021 • Bioinformatics module: Madlung, 2018	• Introductory computing for biologists: http://www.github.com/BILD62 or Libeskind-Hadas and Bush, *Computing for Biologists* (see also Dodds et al., 2012) • "Case studies in Python": https://mark-kramer.github.io/Case-Studies-Python • Advanced coursework: Neuromatch Academy https://compneuro.neuromatch.io

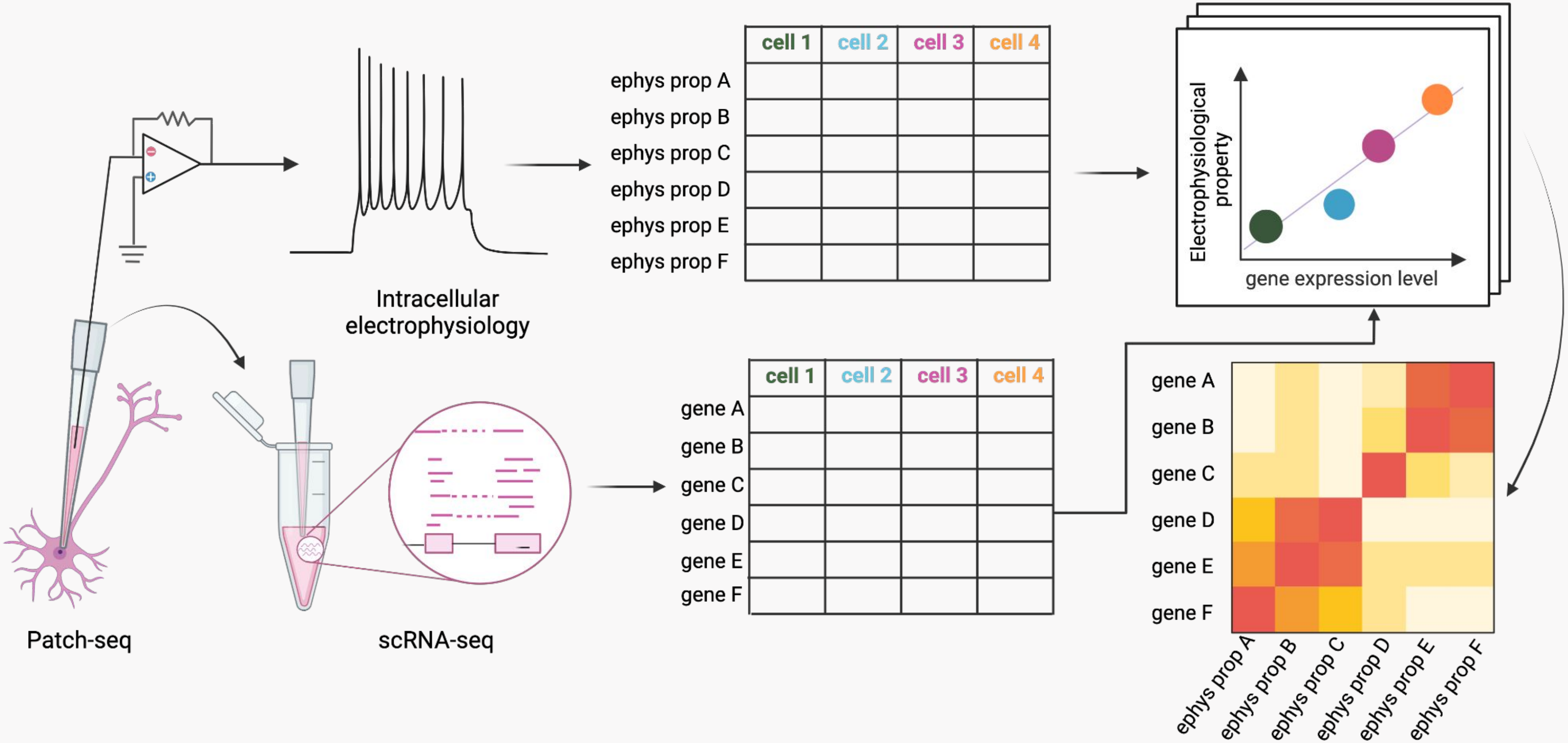
Cell Types Notebook

Introduction & Web Based Lesson:

https://sites.google.com/ucsd.edu/neuroedu/cell-types

Repository: https://github.com/ajuavinett/CellTypesLesson/

PatchSeq Notebook (already in our repo!)



For introductory biology or neuroscience courses

- Analyze & visualize real data from the kinds of biological phenomena you're discussing or from data you're collecting
  - e.g. https://bipn145.github.io/
  - creating plots you can't make with google sheets or excel (e.g., dot plots!)
- Develop simulations to test phenomena
  - E.g. quantal transmission at synapses (Colab Notebook)

To get here, you'll need to reverse engineer what you need students to know!

For my lab class, it's one introductory notebook (1 hour): https://bipn145.github.io/Python/JupyterNotebooks.html

Assessing programming work in your course

- **For completion** — students can submit a PDF or HTML version of their notebook, or the plots they've generated
- **Using reflections** — ask students what they've learned, what surprised them
- **Using an autograder**
  - GradeScope: Grading a Programming Assignment – Gradescope Guides
  - `nbgrader`: https://nbgrader.readthedocs.io/

How to use GitHub

- You already have an account!
- Create a repo for your class or activity
- You can treat it like a file storage system
- *Or* you can get into version control (admittedly, a little messy!)
    - https://voyteklab.com/git/git-primer/
- You can `git clone` or download the repository

Storing & sharing notebooks in Google Drive vs. GitHub

Google Drive

- Light version control (File > Revision History)
- Students are very familiar with it
- Store notebooks alongside files
- A bit harder to find/share OERs
- Cannot build JupyterBook

GitHub

- Enables proper version control
- Students are likely *not* familiar with it
- Need to upload / wget data to Colab
- Easier to find/share OERs
- Easy to build JupyterBook

JupyterBooks allow you to "stitch together" different notebooks

A lab example: https://bipn145.github.io/

Where the code lives: https://github.com/BIPN145/BIPN145.github.io

Another example: https://mark-kramer.github.io/

Yet another example: https://naturalistic-data.org/

More info: https://jupyterbook.org/

Stuff that might be troublesome
(that I'm sure you can handle!)

- Colab possibly updating / changing package versions, which can cause conflicts
  - every module has ***dependencies***
  - **For containers**, you can have what is called a [`requirements.txt`]() file
  - **In Colab**, you can manually uninstall/install packages:

```
!pip uninstall notworkingpackage

!pip install workingpackage
```

Stuff that might be troublesome
(that I'm sure you can handle!)

- Large data that pushes Colab's limits
  - divide into steps
  - handle pre-processing separately
  - move out of Colab
  - delete items from memory

Planning time!

Today's agenda

Breakfast + How to Talk to your Research IT Department (Ashley)	9:00 AM - 9:30 AM
Resources for creating your own Jupyter notebooks (Ashley)	9:30 AM - 10:00 AM
Working Time (Planning & Next Steps)	10:00 AM - 12:00 PM
Lunch	12:00 PM - 1:00 PM
Working Time (Planning & Next Steps)	1:00 PM - 2:30 PM
Wrap up & farewell!	2:30 PM - 3:00 PM

Next Steps

[Next Steps Google Doc]

Step 1

Add what you think you want to do in the next year with what you've learned (lots of options!!)

Step 2

Add you name under your idea

Step 3

Add your name under other ideas you might be interested in

Sharing your open access programming resources

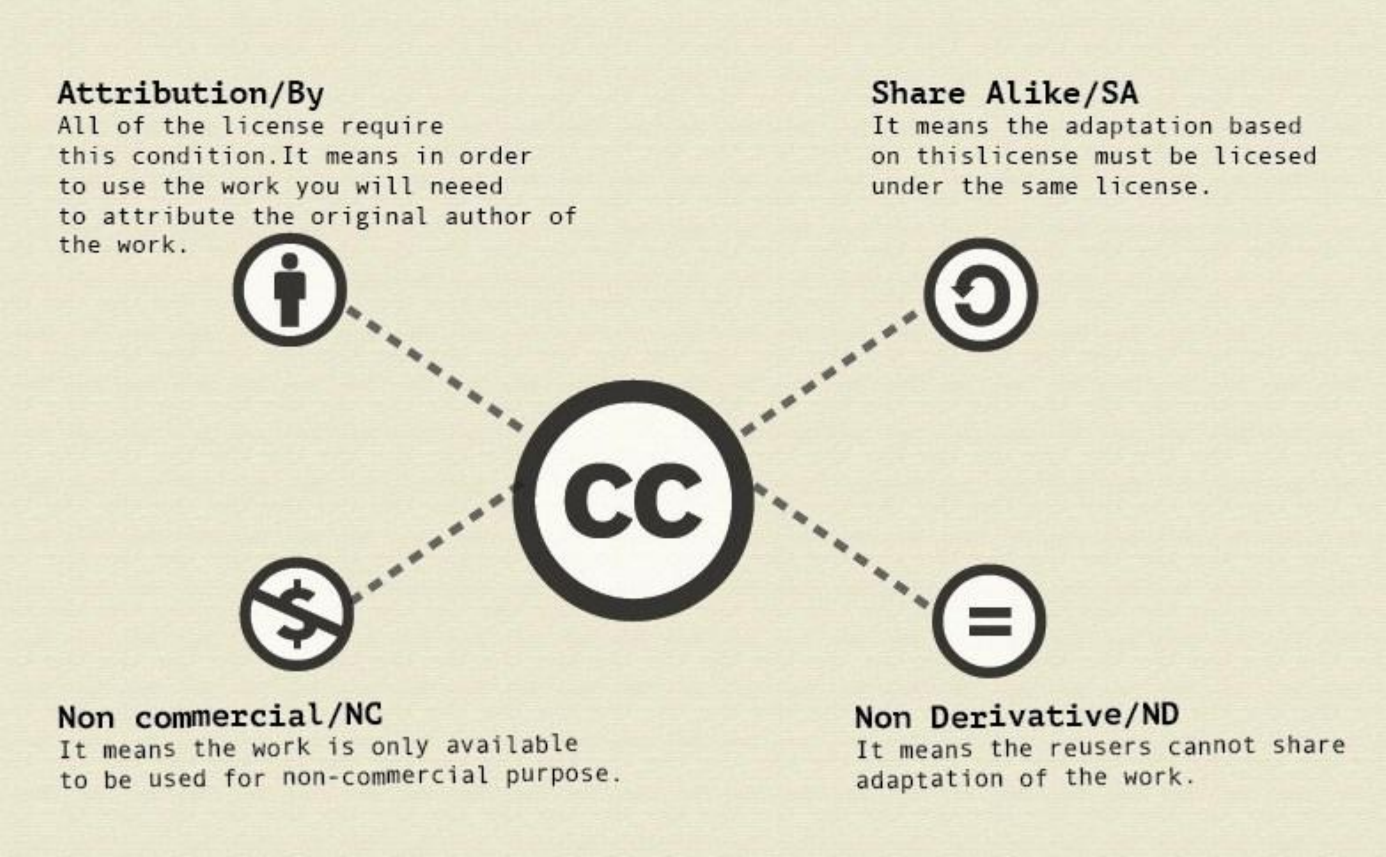
Licenses

You choose:

- Whether or not people can use information commercially
- Whether or not people need to share under the same license
- Example licenses:
  - https://github.com/AllenInstitute/EducationAndEngagement/blob/main/LICENSE
  - https://github.com/ajuavinett/Allen2025/blob/main/LICENSE.txt
- GitHub makes it very easy: https://docs.github.com/en/communities/setting-up-your-project-for-healthy-contributions/adding-a-license-to-a-repository

Creative Commons

- International nonprofit
- These licenses allow creative works to be shared freely on the internet without violating copyright laws



Anatomy of a CC License *by Rie Namba, licensed* *CC BY 4.0*

Creative Commons

The license we use for our OERs

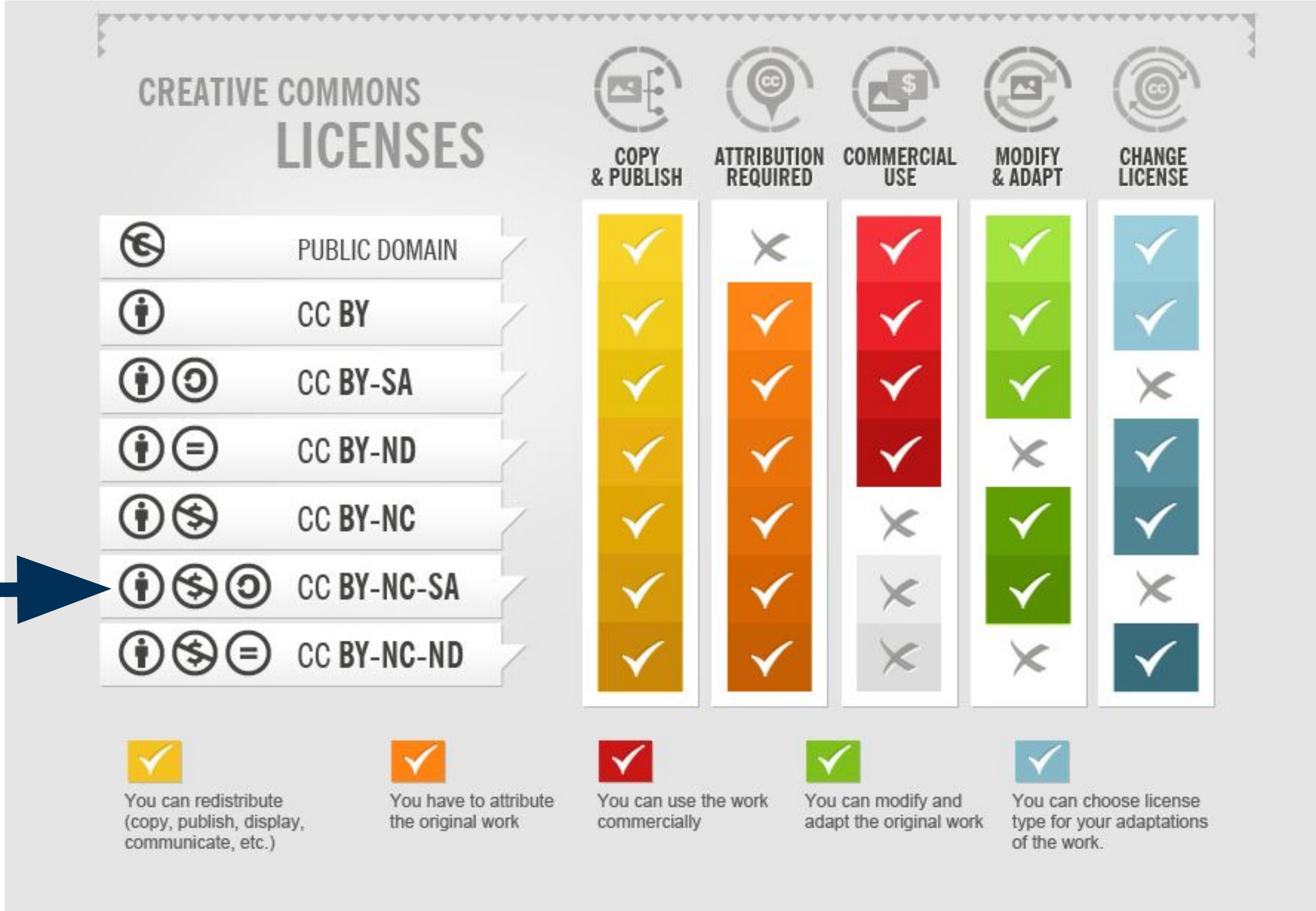


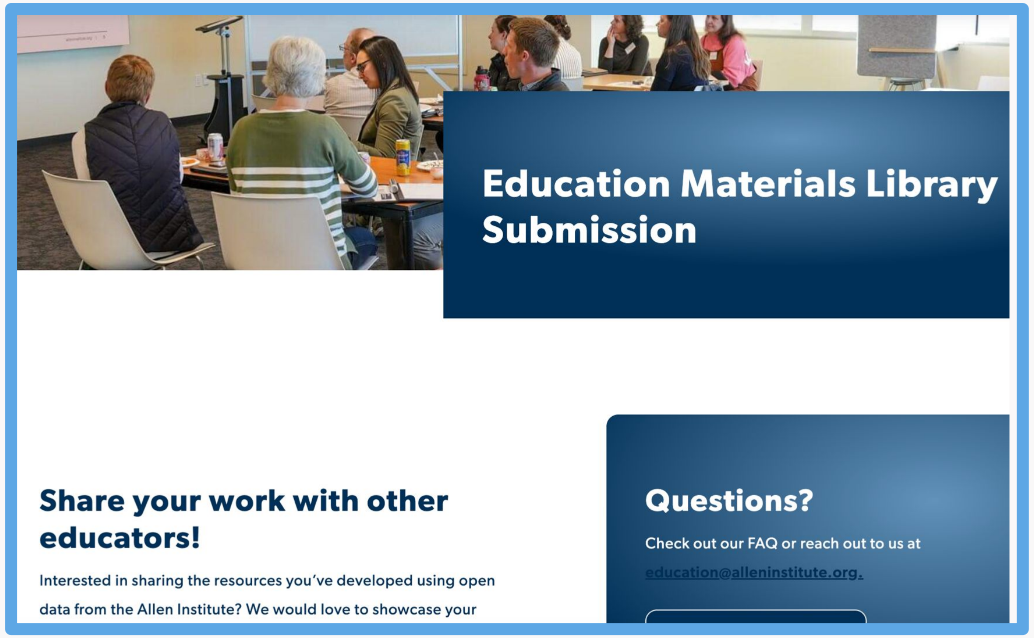
Diagram from [How To Attribute Creative Commons Photos](#) from Foter, licensed [CC BY-SA 3.0](#)

Speaking of publishing, where can you publish OERs that you develop?

- *Journal of Undergraduate Neuroscience Education (JUNE)*
- *CourseSource*
- *Journal of Open Source Education (JOSE)*
- *QUBES*
- *Others…?*

Education materials library submissions

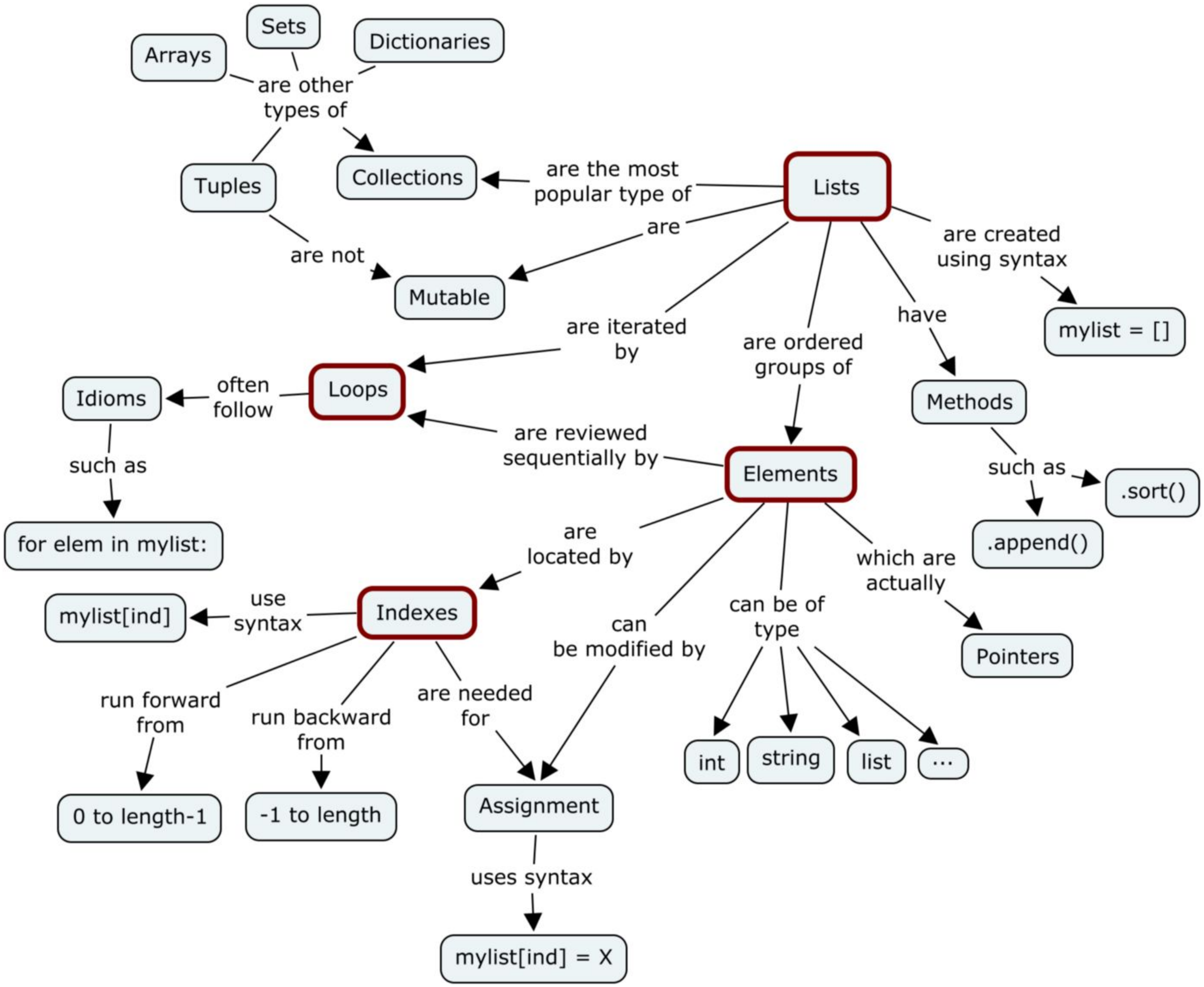
- Members of our educator network can share their work openly via the materials library
- All resources will have **CC BY-NC-SA 4.0** licenses & educators retain the rights to publish elsewhere, etc.
- Living webpages where we can add collaboration requests, student outcome data, and links to publications



This fall: Materials from Teachstravaganza 2024!

Next fall (or later): yours!

Extending what you've seen and learned here



[Mutable vs Immutable Objects in Python | by megha mohan | Medium]

MAKE A PROGRAMMING MEME!

Meme Generator - Imgflip

Add to this slide deck!

Original tweet

Gain more coding-specific pedagogy

The Carpentries Instructor Training https://carpentries.org/instructor-training/

You can also submit lessons to their incubator:
https://carpentries-incubator.org/

Resources from UCB Data8

- The Data8 Pedagogy Guide
- Data 8 Adoption Website
- Online Textbook - Computational and Inferential Thinking: The Foundations of Data Science
- Example course website: https://www.data8.org/materials-sp22/demo.html
- YouTube videos

Data 8 Demo Class(Based off of UCB Spring 2022 materials) Resources ▾

Example Calendar

This page offers links to specific notebooks in the following notebook rendering platforms: JupyterHub, Colab, and CodeSpace. See the Demo Hub, Colab, and CodeSpace columns in the table below. The HTML Notebooks column offers a non-interactive version of the notebook for quick viewing.

You can also view the entire repository of notebooks (rather than one at a time) on these platforms:

- **Jupyterlite (or Jupyterlite(No Footprint)):** This is a browser-based, serverless, no authentication needed instance of JupyterLab. The no-footprint version of these notebooks loads external data files and images from a public url as opposed to storing the external files in the repository.
- **CodeSpace:** This environment is offered via GitHub and can be accessed here. The README file explains in detail how to use this environment. Individual links to each assignment in the table below will open a CodeSpace as well as open the notebook.

Reference Notebooks: DataScience To Pandas

Slides and Discussion/Lab Worksheets: These are all located in the private repository for data 8. If you need access please contact: sean.smorris@berkeley.edu

Week	Lecture Number	Date	Topic	Lecture	Reading	Lab Worksheets	Lab/HW/Project Title	HTML Notebooks	Demo Hub	Colab	CodeSpace
Week 1	1	Wed 01/19	Introduction	Slides, Demos, Video	1.1, 1.2, 1.3	Lab 01 Worksheet	Lab 01: Expressions	Lab 01	Lab 01	Lab 01	Lab 01
	2	Fri 01/21	Cause and Effect	Slides, Video	2		Homework 01 (Due Thu 01/27)	HW 01	HW 01	HW 01	HW 01
Week 2	3	Mon 01/24	Tables	Slides, Demos, Video	3	Lab 02 Worksheet	Lab 02: Table Operations	Lab 02	Lab 02	Lab 02	Lab 02
	4	Wed 01/26	Data Types	Slides, Demos, Video	4, 5						
	5	Fri 01/28	Building Tables	Slides, Demos, Video	6.1, 6.2		Homework 02 (Due Thu 02/03)	HW 02	HW 02	HW 02	HW 02
Week 3	6	Mon 01/31	Census	Slides, Demos, Video	6.3, 6.4	Lab 03 Worksheet	Lab 03: Data Types and Creating & Extending Tables	Lab 03	Lab 03	Lab 03	Lab 03
	7	Wed 02/02	Charts	Slides, Demos, Video	7, 7.1						
	8	Fri 02/04	Histograms	Slides, Demos, Video	7.2, 7.3		Homework 03 (Due Thu 02/10)	HW 03	HW 03	HW 03	HW 03
Week 4	9	Mon 02/07	Functions	Slides, Demos, Video	8, 8.1	Lab 04 Worksheet	Lab 04: Functions and Visualizations	Lab 04	Lab 04	Lab 04	Lab 04

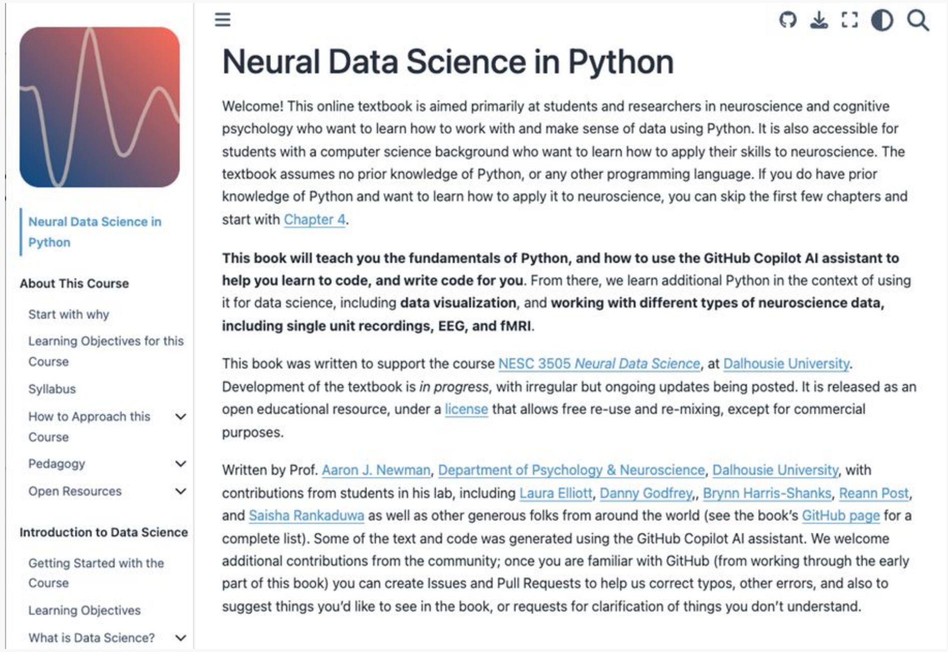
Slide: Theresa McKim

How to continue learning
(introductory materials)

- **Software Carpentries**:
  - Programming with Python
  - Plotting and Programming in Python
- Dataquest (Python Data Science Path)
- https://pyforneuro.com/
- **Videos**: https://www.youtube.com/@OCCTIVE
- CS50: Introduction to Computer Science | Harvard University
  - Full online course in computer science
- https://github.com/Columbia-Neuropythonistas/IntroPythonForNeuroscientists2023

How to continue learning
(more advanced materials)

- Open Education Resources – Neuromatch
  - More advanced courses,
    not just neuroscience!
- Computational and Inferential Thinking: The Foundations of Data Science
  - Another JupyterBook!
    For Data8 @ Berkeley.



Neural Data Science in Python

About This Course

Start with why

Learning Objectives for this Course

Syllabus

How to Approach this Course

Pedagogy

Open Resources

Introduction to Data Science

Getting Started with the Course

Learning Objectives

What is Data Science?

Neural Data Science in Python

Welcome! This online textbook is aimed primarily at students and researchers in neuroscience and cognitive psychology who want to learn how to work with and make sense of data using Python. It is also accessible for students with a computer science background who want to learn how to apply their skills to neuroscience. The textbook assumes no prior knowledge of Python, or any other programming language. If you do have prior knowledge of Python and want to learn how to apply it to neuroscience, you can skip the first few chapters and start with Chapter 4.

This book will teach you the fundamentals of Python, and how to use the GitHub Copilot AI assistant to help you learn to code, and write code for you. From there, we learn additional Python in the context of using it for data science, including **data visualization**, and **working with different types of neuroscience data, including single unit recordings, EEG, and fMRI.**

This book was written to support the course NESC 3505 *Neural Data Science*, at Dalhousie University. Development of the textbook is *in progress*, with irregular but ongoing updates being posted. It is released as an open educational resource, under a license that allows free re-use and re-mixing, except for commercial purposes.

Written by Prof. Aaron J. Newman, Department of Psychology & Neuroscience, Dalhousie University, with contributions from students in his lab, including Laura Elliott, Danny Godfrey,, Brynn Harris-Shanks, Reann Post, and Saisha Rankaduwa as well as other generous folks from around the world (see the book's GitHub page for a complete list). Some of the text and code was generated using the GitHub Copilot AI assistant. We welcome additional contributions from the community; once you are familiar with GitHub (from working through the early part of this book) you can create Issues and Pull Requests to help us correct typos, other errors, and also to suggest things you'd like to see in the book, or requests for clarification of things you don't understand.

https://neuraldatascience.io/intro.html

- Case-Studies-Python — Case Studies in Neural Data Analysis
- https://nwb4edu.github.io/ — work with Neurodata Without Borders!
- Classes at your institution?
- More than just coding:
  https://training.incf.org/courses-and-collections

How to ask questions

About Allen Data: https://community.brain-map.org/

About programming: the internet or a chatbot, probably!

About lesson plans: the authors!

Data Science Slack Communities

https://dslc.io/

https://datascienceeducation.substack.com/

Biological and Environmental Data Education (BEDE) Network: ask to join Slack channel